

GAL Seminar 4

ex 1

Fie B baza canonică din $\mathbb{R}^3$ și

$$B_1 = \{ (1, 1, 1), (2, 2, 3), (2, 1, 5) \}.$$

Determinați matricea de trecere $B \xrightarrow{\quad} B_1$

ex 2

$$\text{Fie } B_1 = \{ (1, 1, 1), (2, 2, 3), (2, 1, 5) \}$$

$$B_2 = \{ (1, 1, 0), (1, 0, 1), (0, 1, 1) \}$$

Încetă cele două matrici de trecere. $B_1 \xrightarrow{\quad} B_2$

ex 3

$$\text{Fie } L = \{ x \in \mathbb{R}^3 \mid x_1 - 2x_2 - 3x_3 = 0 \}$$

a) Det. 2 baze din L

b) Det. mat. de trecere $B_1 \xrightleftharpoons[A^{-1}]{A} B_2$

Aplicații liniare

$$\text{Ker } f = \{ x \mid f(x) = 0 \}$$

$$\text{Im } f = \{ y \in W \mid \exists x \in V \quad f(x) = y \}$$

ex 4

$$\text{Fie } f: \mathbb{R}^3 \rightarrow \mathbb{R}^4$$

$$f(x) = (x_1 + x_2 - x_3, \quad x_1 - x_2 + x_3, \quad 2x_1 - x_2, \quad x_1 - 3x_3)$$

a) Arătați că f este ap. liniară

b) Găsiți câte o bază în Ker și $\text{Im } f$

ex 5

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad f(x) = (x_1 + x_2 - x_3, \quad x_1 - x_2 + x_3)$$

$\text{Ker } f, \text{Im } f$, câte o bază

ex 1

Fie B baza canonică din $\mathbb{R}^3$ și

$$B_1 = \{ (1, 1, 1), (2, 2, 3), (2, 1, 5) \}.$$

Determinați matricea de trecere $B \xrightarrow{\quad} B_1$

Sol:

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 5 \end{pmatrix}$$

$$B_1 \xrightarrow[A^{-1}]{A} B$$

$$A^{-1} = \left(\begin{array}{ccc|ccc} \boxed{1} & 2 & 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 3 & 5 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\begin{array}{l} L_2 - L_1 \\ L_3 - L_1 \end{array}]{L_2 - L_1}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & \boxed{1} & 3 & -1 & 0 & 1 \end{array} \right) \xrightarrow[\longrightarrow]{L_2 \leftrightarrow L_3}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & \boxed{1} & 3 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{array} \right) \xrightarrow{\longrightarrow} \begin{array}{l} L_1 - 2L_2 \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -4 & 3 & 0 & -2 \\ 0 & 1 & 3 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array} \right) \quad \begin{array}{l} L_1 + 4L_3 \\ \hline L_2 - 3L_3 \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -4 & -2 \\ 0 & 1 & 0 & -4 & 3 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 7 & -4 & -2 \\ -4 & 3 & 1 \\ 1 & -1 & 0 \end{pmatrix} \quad x = (1, 1, 1)$$

$$[x]_{B_1} = A^{-1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 7 & -4 & -2 \\ -4 & 3 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x = 1 \cdot (1, 1, 1) + 0 \cdot (2, 2, 3) + 0 \cdot (2, 1, 5)$$

ex 2

$$\text{Fie } B_1 = \{ (1, 1, 1), (2, 2, 3), (2, 1, 5) \}$$

$$B_2 = \{ (1, 1, 0), (1, 0, 1), (0, 1, 1) \}$$

Scriseti cele doua matrici de trecere. $B_1 \xrightarrow{\quad} B_2$

Sol:

$$A^{-1} = \begin{pmatrix} 7 & -4 & -2 \\ -4 & 3 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

$$B_1 \xrightarrow{A} B \leftarrow B_2$$

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 5 \end{pmatrix}$$

$$B_1 \xrightarrow{B^{-1} \cdot A} B_2$$

$$A^{-1} B$$

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$A^{-1} B = \begin{pmatrix} 7 & -4 & -2 \\ -4 & 3 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 5 & -6 \\ -1 & -3 & 4 \\ 0 & 1 & -1 \end{pmatrix}$$

$$B^{-1} A = \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 3 & 5 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 2 & 2 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 3 & 5 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 2 & 1 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 2 & 1 & 3 & 4 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{3}{2} & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -1 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} & 3 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{3}{2} & 2 \end{array} \right)$$

ex 3

$$\text{Fix } L = \{ x \in \mathbb{R}^3 \mid x_1 - 2x_2 - 3x_3 = 0 \}$$

a) Det. 2 base $\dim L$

Sol:

$$x_1 - 2x_2 - 3x_3 = 0$$

$$x_1 = 2x_2 + 3x_3$$

$$L = \{ (2x_2 + 3x_3, x_2, x_3) \mid x_2, x_3 \in \mathbb{R} \}$$

$$L = \langle (2, 1, 0), (3, 0, 1) \rangle$$

$$B_1 = \{ f_1, f_2 \} = \{ (2, 1, 0), (3, 0, 1) \}$$

$$B_2 = \{ g_1, g_2 \} = \{ (5, 1, 1), (1, -1, 1) \}$$

b) Det. mat. de transition $B_1 \xrightleftharpoons[A^{-1}]{A} B_2$

$$\text{Pt } A = \left(\begin{array}{cc|cc} 1 & 5 & 2 & 3 \\ -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) = \left(\begin{array}{cc|cc} 1 & 5 & 2 & 3 \\ 0 & 6 & 3 & 3 \\ 0 & -4 & -2 & -2 \end{array} \right)$$

$$= \left(\begin{array}{cc|cc} 1 & 5 & 2 & 3 \\ 0 & 2 & 1 & 1 \\ 0 & 2 & 1 & 1 \end{array} \right) = \left(\begin{array}{cc|cc} 1 & 5 & 2 & 3 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$= \left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right)$$

$$f_1 = -\frac{1}{2} (1, -1, 1) + \frac{1}{2} (5, 1, 1) = (2, 1, 0)$$

$$A^{-1} = \left(\begin{array}{cc|cc} 2 & 3 & 1 & 5 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right) = \left(\begin{array}{cc|cc} \boxed{1} & 0 & -1 & 1 \\ 2 & 3 & 1 & 5 \\ 0 & 1 & 1 & 1 \end{array} \right)$$

$$= \left(\begin{array}{cc|cc} 1 & 0 & -1 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 1 & 1 & 1 \end{array} \right) = \left(\begin{array}{cc|cc} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right)$$

$$g_2 = (2, 1, 0) + (3, 0, 1) = (5, 1, 1)$$

□

Aplicații liniare

Fie $f: V \rightarrow W$, $f(ax + by) = a \cdot f(x) + b \cdot f(y)$

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, f liniară $\Leftrightarrow \exists A \in M_{n,m}(\mathbb{K})$ a.i.

$$f(x) = A \cdot x$$

$\text{Ker } f = \{ x \in V \mid f(x) = 0 \}$

$\text{Im } f = \{ y \in W \mid \exists x \in V \text{ } f(x) = y \}$

$$\dim \text{Ker } f + \dim \text{Im } f = \dim V$$

ex 4

$$\text{Fie } f: \mathbb{R}^3 \rightarrow \mathbb{R}^4$$

$$f(x) = (x_1 + x_2 - x_3, x_1 - x_2 + x_3, 2x_1 - x_2, x_1 - 3x_3)$$

a) Arătați că f este op. liniară

$$f(x) = A \cdot x$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & -3 \end{pmatrix} \Rightarrow \text{op. liniară}$$

b) Găsiți câte o bază în $\text{Ker } f$ și $\text{Im } f$

$$x \in \text{Ker} \Rightarrow \begin{cases} x_1 + x_2 - x_3 = 0 \\ x_1 - x_2 + x_3 = 0 \\ 2x_1 - x_2 = 0 \\ x_1 - 3x_3 = 0 \end{cases}$$

$$\begin{pmatrix} \boxed{1} & 1 & -1 \\ 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & -3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 2 \\ 0 & -3 & 2 \\ 0 & -1 & -2 \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & \boxed{1} & -1 \\ 0 & -3 & 2 \\ 0 & -1 & -2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -3 \end{pmatrix}$$

$$\text{rang}(A) = 3 \Rightarrow 3 \text{ independente} \Rightarrow$$

$$\text{Ker}(f) = \{0\} \Rightarrow f \text{ inj}$$

Pt. Imf

$$x = x_1 \cdot e_1 + x_2 \cdot e_2 + x_3 \cdot e_3$$

$$f(x) = x_1 \cdot f(e_1) + x_2 \cdot f(e_2) + x_3 \cdot f(e_3)$$

$$\text{Im} f = \langle f(e_1), f(e_2), f(e_3) \rangle$$

$$= \langle (1, 1, 2, 1), (1, -1, -1, 0), (-1, 1, 0, -2) \rangle$$

$$\text{i.e. } (1, 0, 0), (0, 1, 0), (0, 0, 1)$$

$$A = (f(e_1) \mid f(e_2) \mid f(e_3))$$

$$\text{Rang } A = 3 \Rightarrow \{f(e_1) \mid f(e_2) \mid f(e_3)\} \text{ SLI}$$

$$\text{Rang } \mathbb{R}^3 < \mathbb{R}^4 \Rightarrow \text{NU e surj}$$

ex 5

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad f(x) = (x_1 + x_2 - x_3, \quad x_1 - x_2 + x_3)$$

Ker f, Im f, câte & baza

Sol:

$$\text{pt Ker } f \quad \begin{cases} x_1 + x_2 - x_3 = 0 \\ x_1 - x_2 + x_3 = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

$$x_1 = 0$$

$$x_2 - x_3 = 0 \quad \Rightarrow \quad x_2 = x_3$$

$$\text{Ker } f = \langle (0, 1, 1) \rangle$$

Pt $\text{Im } f$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \quad \text{i.e. mat } f$$

$\text{Im } f$ este generat de sol. lini $A \Rightarrow \text{Im } f \subset \text{sol lini } A$

$$\text{Im } f = \langle (1, 1), (1, -1), (-1, 1) \rangle$$

$$\Rightarrow \langle (1, 1), (1, -1) \rangle \Rightarrow \mathbb{R}^2 \text{ este surj}$$

$$\dim \text{Ker } f = 3 - \text{rang}(A) = 3 - 2 = 1$$

$$\dim \text{Im } f = \text{rang } A = 2$$